ANX-PR/CL/001-01
LEARNING GUIDE

SUBJECT

53002010 - IntroducciÓn A Los Mercados EnergÉticos

DEGREE PROGRAMME

05BK - Máster Universitario En Ingeniería De La Energía

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 1

Index

Learning guide

1. Description

1.1. Subject details

Name of the subject	53002010 - Introducción a los Mercados Energéticos
No of credits	3 ECTS
Type	Optional/elective
Academic year ot the programme	First year
Semester of tuition	Semester 1
Tuition period	September-January
Tuition languages	English
Degree programme	05BK - Máster Universitario en Ingeniería de la Energía
Centre	05 - E.T.S. De Ingenieros Industriales
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Carlos Enrique Vazquez Martinez (Subject coordinator)	518	vazquez.martinez@upm.es	Tu - 11:00 - 13:00 W - 11:00 - 13:00 Th - 11:00 - 13:00 Please, request an appointment by email

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Skills and learning outcomes *

3.1. Skills to be learned

CB10 - Que los estudiantes posean las habilidades de aprendizaje que les permitan continuar estudiando de un modo que habrá de ser en gran medida autodirigido o autónomo.

CB7 - Que los estudiantes sepan aplicar los conocimientos adquiridos y su capacidad de resolución de problemas en entornos nuevos o poco conocidos dentro de contextos más amplios (o multidisciplinares) relacionados con su área de estudio

CE13 - Entender la evolución y el funcionamiento de los mercados de petróleo, gas y electricidad. Conocer los principales tipos de diseño de los mercados de electricidad y gas que existen en la experiencia internacional y los criterios bajo los que se han diseñado, y ser capaz de analizar cuál es la regulación más adecuada para cada situación.

CE16 - Aplicar conocimientos y habilidades adquiridas para la práctica profesional de alto nivel y la gestión de equipos en las empresas del sector energético.

CE17 - Comprender los procesos que integran el ciclo de vida de los procesos energéticos, desde la obtención del recurso primario, hasta su desmantelamiento, y su integración en la economía circular.

CE18 - Entender la optimización de costes en una empresa: coste marginal, coste medio, coste hundido, coste de oportunidad, aplicados al sector de la energía. Analizar costes en el sector de la energía.

CE4 - Comprender y aplicar los principios de funcionamiento, formación de precios y equilibrio en los mercados energéticos, tanto en condiciones de competencia perfecta como en condiciones de competencia imperfecta

CE5 - Comprender y conocer las herramientas regulatorias y normativas del sector energético.

CG1 - Aplicar conocimientos de ciencias y tecnologías avanzadas a la práctica profesional o investigadora de la Ingeniería Energética.

CG2 - Poseer capacidad para diseñar, desarrollar, implementar, gestionar y mejorar productos, sistemas y procesos en los distintos ámbitos energéticos, usando técnicas analíticas, computacionales o experimentales avanzadas.

CT1 - Aplica. Habilidad para aplicar conocimientos científicos, matemáticos y tecnológicos en sistemas relacionados con la práctica de la ingeniería.

CT10 - Conoce. Conocimiento de los temas contemporáneos.

CT11 - Usa herramientas. Habilidad para usar las técnicas, destrezas y herramientas ingenieriles modernas necesarias para la práctica de la ingeniería.

CT3 - Diseña. Habilidad para diseñar un sistema, componente o proceso que alcance los requisitos deseados teniendo en cuenta restricciones realistas tales como las económicas, medioambientales, sociales, políticas, éticas, de salud y seguridad, de fabricación y de sostenibilidad.

CT5 - Resuelve. Habilidad para identificar, formular y resolver problemas de ingeniería.

CT9 - Se actualiza. Reconocimiento de la necesidad y la habilidad para comprometerse al aprendizaje continuo.

3.2. Learning outcomes

RA164 - Entender la evolución y el funcionamiento de los mercados de petróleo

RA136 - Conocer la situación actual, evolución hasta el momento y perspectivas futuras de los diferentes mercados de energía

RA168 - Analizar y entender los diferentes tipos de diseño de mercado de los mecanismos de apoyo a la inversión de generación eléctrica, convencional y renovable

RA167 - Analizar y entender los diferentes tipos de diseño de los mercados de electricidad, tanto de energía como de servicios complementarios

RA165 - Comprender la situación actual y los diferentes tipos de diseño de los mercados de gas

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

4. Brief description of the subject and syllabus

4.1. Brief description of the subject

The course focuses on the study of the main energy markets and their regulation. The oil, natural gas, and electricity markets will be addressed. Each topic will cover at least two types of analysis. First, a description of the economic characteristics of the sectors and their implications for the behavior of stakeholders. This includes a review of the current situation of the sector, its historical evolution, and future prospects. Second, a description of the applicable regulations and market design. The emphasis is not so much on the specific standards operating in the different markets, but rather on their foundations, design criteria, and the alternatives that could have been applied at each point. In general, emphasis will be placed on an international analysis of the markets, covering different implementation alternatives that have been adopted worldwide.

4.2. Syllabus

1. Oil

2. Natural gas

    2.1. Fundamentals

    2.2. Regulation of access and investment: UK vs USA models

3. Spot power markets

    3.1. Energy markets

    3.2. Congestion management and losses

    3.3. Operational reserves

5. Schedule

5.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	**Introduction** Duration: 01:00 Lecture **Oil markets** Duration: 01:00 Lecture			
2	**Follow-up test** Duration: 00:05 Additional activities **Oil markets** Duration: 01:55 Lecture			**Follow-up test** Written test Progressive assessment Presential Duration: 00:05
3	**Follow-up test** Duration: 00:05 Additional activities **Natural gas markets** Duration: 01:55 Lecture			**Follow-up test** Written test Progressive assessment Presential Duration: 00:05
4	**Follow-up test** Duration: 00:05 Additional activities **Natural gas markets** Duration: 01:55 Lecture			**Follow-up test** Written test Progressive assessment Presential Duration: 00:05
5	**Follow-up test** Duration: 00:05 Additional activities **Natural gas markets** Duration: 01:55 Lecture			**Follow-up test** Written test Progressive assessment Presential Duration: 00:05
6	**midterm exam: oil & gas** Duration: 00:50 Additional activities **Electricity spot markets: energy** Duration: 01:10 Lecture			**Mid term exam** Written test Progressive assessment Presential Duration: 00:50

7	**Follow-up test** Duration: 00:05 Additional activities **Electricity spot markets: energy** Duration: 01:55 Lecture			**Follow-up test** Written test Progressive assessment Presential Duration: 00:05
8	**Follow-up test** Duration: 00:05 Additional activities **Electricity spot markets: energy** Duration: 01:55 Lecture			**Follow-up test** Written test Progressive assessment Presential Duration: 00:05
9	**Follow-up test** Duration: 00:05 Additional activities **Electricity spot markets: energy** Duration: 01:55 Lecture			**Follow-up test** Written test Progressive assessment Presential Duration: 00:05
10	**midterm exam: day-ahead markets** Duration: 00:50 Additional activities **Electricity spot markets: congestion management and losses** Duration: 01:10 Lecture			**Mid term exam** Written test Progressive assessment Presential Duration: 00:50
11	**Follow-up test** Duration: 00:05 Additional activities **Electricity spot markets: congestion management and losses** Duration: 01:55 Lecture			**Follow-up test** Written test Progressive assessment Presential Duration: 00:05
12	**Follow-up test** Duration: 00:05 Additional activities **Electricity spot markets: congestion management and losses** Duration: 01:55 Lecture			**Follow-up test** Written test Progressive assessment Presential Duration: 00:05
13	**Follow-up test** Duration: 00:05 Additional activities **Electricity spot markets: operational reserves** Duration: 01:55 Lecture			**Follow-up test** Written test Progressive assessment Presential Duration: 00:05
14	**Electricity spot markets: operational reserves** Duration: 01:10 Lecture **midterm exam: congestion management and operational reserves** Duration: 00:50			**Mid term exam** Written test Progressive assessment Presential Duration: 00:50

	Additional activities			
15				
16				
17				**Final exam** Written test Progressive assessment Presential Duration: 02:00 **Final exam** Written test Global examination Presential Duration: 02:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

6. Activities and assessment criteria

6.1. Assessment activities

6.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
2	Follow-up test	Written test	Face-to-face	00:05	2.5%	0 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT5 CT9 CT10 CT11 CE4 CE5 CE13 CE16 CE17 CE18
3	Follow-up test	Written test	Face-to-face	00:05	2.5%	0 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT5 CT9 CT10 CT11 CE4 CE5 CE13 CE16 CE17 CE18
							CB7 CB10 CG1 CG2 CT1 CT3 CT5 CT9

4	Follow-up test	Written test	Face-to-face	00:05	2.5%	0 / 10	CT10 CT11 CE4 CE5 CE13 CE16 CE17 CE18
5	Follow-up test	Written test	Face-to-face	00:05	2.5%	0 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT5 CT9 CT10 CT11 CE4 CE5 CE13 CE16 CE17 CE18
6	Mid term exam	Written test	Face-to-face	00:50	25%	0 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT5 CT9 CT10 CT11 CE4 CE5 CE13 CE16 CE17 CE18
7	Follow-up test	Written test	Face-to-face	00:05	2.5%	0 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT5 CT9 CT10 CT11 CE4 CE5 CE13 CE16

							CE17 CE18
8	Follow-up test	Written test	Face-to-face	00:05	2.5%	0 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT5 CT9 CT10 CT11 CE4 CE5 CE13 CE16 CE17 CE18
9	Follow-up test	Written test	Face-to-face	00:05	2.5%	0 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT5 CT9 CT10 CT11 CE4 CE5 CE13 CE16 CE17 CE18
10	Mid term exam	Written test	Face-to-face	00:50	25%	0 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT5 CT9 CT10 CT11 CE4 CE5 CE13 CE16 CE17 CE18

11	Follow-up test	Written test	Face-to-face	00:05	2.5%	0 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT5 CT9 CT10 CT11 CE4 CE5 CE13 CE16 CE17 CE18
12	Follow-up test	Written test	Face-to-face	00:05	2.5%	0 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT5 CT9 CT10 CT11 CE4 CE5 CE13 CE16 CE17 CE18
13	Follow-up test	Written test	Face-to-face	00:05	2.5%	0 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT5 CT9 CT10 CT11 CE4 CE5 CE13 CE16 CE17 CE18

14	Mid term exam	Written test	Face-to-face	00:50	25%	0 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT5 CT9 CT10 CT11 CE4 CE5 CE13 CE16 CE17 CE18
17	Final exam	Written test	Face-to-face	02:00	75%	5 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT5 CT9 CT10 CT11 CE4 CE5 CE13 CE16 CE17 CE18

6.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
17	Final exam	Written test	Face-to-face	02:00	100%	5 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT5 CT9 CT10 CT11 CE4 CE5 CE13 CE16 CE17 CE18

6.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Extraordinary exam	Written test	Face-to-face	02:00	100%	5 / 10	CB7 CB10 CG1 CG2 CT1 CT3 CT5 CT9 CT10 CT11 CE4 CE5 CE13 CE16 CE17 CE18

6.2. Assessment criteria

For those students following the continuos assesment of the course,

- 25% of the final grade corresponds to the follow-up tests. There is no minimum grade for each individual test, but an average grade of 4 is required. Also, it is required to take at least 70% of the follow-up tests. Otherwise, the student will have to take the whole final exam.

- 75% of the final grade corresponds to the three midterm exams, each of them weighting 25%. A minimum grade of 5 is required in each of the midterm exams. In case any of the midterm exams fails under that minumum grade, the topics corresponding to that midterm exam will have to be evaluated again at the final exam. The topics covered by midterm exams with grades over 5 will not have to be evaluated again at the final exam. The relevant grade for calculating the global average will be the last exam taken, for each part.

For students taking the course under the "final exam only" option,

- 100% of their final grade will be awarded for the final exam

When an extraordinary exam is required, the entire syllabus must be evaluated, and 100% of the grade will be awarded for the extraordinary exam.

7. Teaching resources

7.1. Teaching resources for the subject

Name	Type	Notes
D. Yergin, "The Prize: The Epic Quest for Oil, Money and Power", 2009	Bibliography	Oil markets. Descriptive, not too much theory
https://www.oxfordenergy.org/	Web resource	Gas markets
S. Stoft, "Power System Economics: Designing Markets for Electricity", 2002	Bibliography	Power markets, focus on market design

I.J. Pérez-Arriaga (Editor), "Regulation of the Power Sector", Springer, 2013	Bibliography	Power markets, focus on market design
S. Hunt, "Making Competition Work in Electricity", 2002	Bibliography	Power markets, focus on competition
http://www.irena.org	Web resource	Renewable energy. Focus on costs, not on marktes

8. Other information

8.1. Other information about the subject

This course is related to Sustainable Development Goal number 7: affordable and clean energy